

ABSTRACT

This project focuses on the design and implementation of a dual-output transformer-based regulated DC power supply capable of delivering 9V and 12V regulated outputs. A 12-0-12V, 1A center-tapped step-down transformer is used to convert the 230V AC mains into a lower AC voltage suitable for rectification. The AC output from the transformer is fed into a bridge rectifier constructed using four 1N4007 diodes, which efficiently converts AC into pulsating DC. A high-value filter capacitor is placed at the rectifier output to smoothen voltage ripples and provide a stable DC level. To achieve precise voltage regulation, two linear regulator ICs—LM7809 and LM7812—are individually connected to the filtered DC supply. These regulators provide fixed and stable 9V and 12V outputs, protecting the load from fluctuations and ensuring constant performance under varying input conditions. For indication and safety, each regulator output is connected to an LED through a 1 k Ω resistor, allowing visual monitoring of proper voltage regulation and circuit operation.

The designed supply offers excellent characteristics such as low ripple, thermal shutdown protection, short-circuit safety, and stable regulated outputs, making it suitable for testing electronic circuits, embedded systems, communication modules, and general laboratory use. This project successfully demonstrates the practical implementation of a reliable, low-cost power supply using fundamental electronic components, reinforcing the understanding of rectification, filtering, and linear voltage regulation.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
DC	Direct Current
AC	Alternating Current
IC	Integrated Circuit
LED	Light Emitting Diode
PCB	Printed Circuit Board
CT	Center Tap (Transformer)
PIV	Peak Inverse Voltage
mA	Milliampere
VA	Volt-Ampere (Transformer rating)
LM7809	9V Voltage Regulator IC
LM7812	12V Voltage Regulator IC
1N4007	Silicon Rectifier Diode
RMS	Root Mean Square
DC Load	Direct Current Load
SMPS	Switched Mode Power Supply
RF	Radio Frequency

LIST OF SYMBOLS

Symbol	Meaning
AC	Alternating Current
DC	Direct Current
V	Voltage
I	Current
A	Ampere
Hz	Hertz (frequency)
μF / μF	Microfarad (capacitor unit)
V _{rms}	RMS Voltage
V _{peak}	Peak Voltage
V _{DC}	Direct Current Voltage
V _{out}	Output Voltage
V _{in}	Input Voltage
Ω (Ohm)	Resistance unit
k Ω	Kilo-ohm
°C	Temperature in Celsius
W	Watt (Power)
V _p	Primary Voltage
V _s	Secondary Voltage
N _p	Number of Primary Turns
N _s	Number of Secondary Turns
f	Frequency
ΔV (Delta V)	Ripple Voltage

CHAPTER 1 : INTRODUCTION

Electronic systems are an essential part of modern life, and almost every electronic device requires a stable, smooth and regulated DC power source for its proper functioning. Whether it is a small LED indicator, a microcontroller, a communication device, an amplifier, or a sensor, all require a specific DC voltage that must remain constant irrespective of fluctuations in the input power.

The purpose of a regulated power supply is to ensure that the output remains stable, noise-free, and safe for all types of electronic components.

A regulated power supply converts high-voltage AC (Alternating Current) from the mains into low-voltage DC (Direct Current). This process involves several stages such as stepping down the voltage, rectification, filtering, and regulation. Each stage plays a crucial role in ensuring that the output is reliable and suitable for use in electronic circuits.

This chapter provides an in-depth explanation of each building block of a transformer-based regulated power supply with dual output (9V & 12V). It explains the working principles, importance, and purpose of each component in detail.

1.1 INTRODUCTION TO POWER SUPPLY

A power supply is a fundamental building block of any electronic or electrical system. No electronic

circuit can function without a reliable source of power. A power supply's main purpose is to convert electrical energy from one form to another in such a way that electronic components receive a stable, noise-free, and regulated voltage.

In our households, we commonly receive 230V AC power from the main grid. However, almost all modern electronic devices operate on constant DC voltages such as 5V, 9V, 12V, or 15V. These DC voltages remain fixed and do not change their polarity or magnitude with time. On the other hand, AC voltage continuously changes its polarity and magnitude (50 times per second in India). If such fluctuating voltage is applied directly to electronic components, they will immediately burn or malfunction.

Therefore, the main functions of a power supply are:

- To convert AC → DC
- To reduce high AC voltage to a safer level
- To remove ripples and provide smooth output
- To regulate the voltage so it remains constant
- To protect circuits from spikes, noise, and surges

A properly designed power supply ensures:

- Long life of electronic components
- High efficiency
- Stable operation
- No damage due to voltage fluctuation

Every small or large electronic device, such as mobile chargers, televisions, radios, communication devices, microcontroller projects, and robotics systems, use power supplies in some form.

1.2 TRANSFORMER (STEP-UP & STEP-DOWN)

A transformer is an AC electrical device that works on the principle of mutual electromagnetic induction. It consists of two main coils: the Primary coil and the Secondary coil, wound around a laminated iron core.

How a transformer works

When AC voltage is applied to the primary coil, an alternating magnetic flux is created in the core. This alternating flux induces a voltage in the secondary coil. The induced voltage depends on the number of turns of the coils.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Where,

V_s = Secondary voltage

V_p = Primary voltage

N_s, N_p = Number of turns

Types of Transformers

1. Step-Up Transformer

- Increases AC voltage
- Secondary coil turns are more than primary
- Used in power transmission, inverters, and microwave ovens

Example:

120V AC → 240V AC

2. Step-Down Transformer

Decreases AC voltage

Secondary turns are less

Used in chargers, power supplies, adapters

Example:

230V AC → 12V AC (used in this project)

Advantages of Using a Transformer

- Provides electrical isolation
- Safety from high voltage
- Reduces heating effects
- Highly efficient (up to 98%)
- No moving parts (long life)

Transformer Used in This Project

A 12-0-12V, 1A Center-Tapped Step-Down Transformer is used.

Why 12-0-12V?

It provides:

- Two 12V AC outputs
- One common ground (0V)

This helps in designing dual-voltage circuits and is commonly used in amplifiers and regulated power supplies.

Output after rectification:

12V AC → ~ 16.8V DC (peak)

This is ideal for 7809 & 7812 regulators.

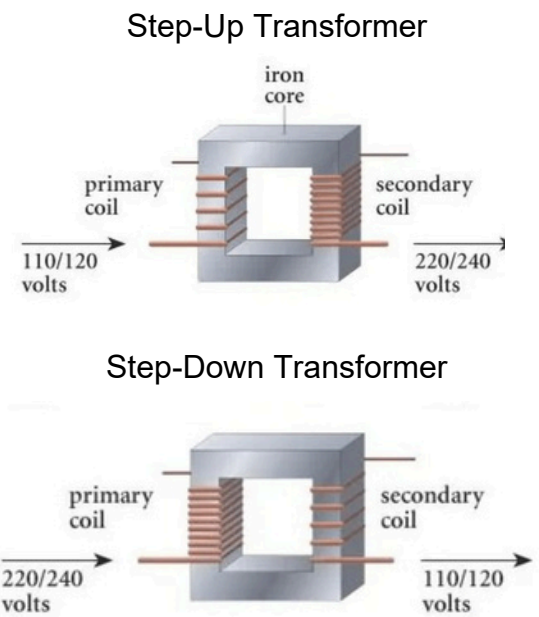


Fig 1.2.1

1.3 BRIDGE RECTIFIER

A rectifier converts Alternating Current (AC) into Direct Current (DC).

The most commonly used configuration in power supplies is the Bridge Rectifier, which uses four diodes arranged in a special pattern.

In this project, IN4007 diodes are used, which can withstand:

- 1000V peak inverse voltage
- 1A continuous current

Working of Bridge Rectifier (Detailed Explanation)

Positive Half Cycle

- Two diodes conduct (D1 & D2)
- Current flows through the load in one direction
- Output is positive

Negative Half Cycle

- Other two diodes conduct (D3 & D4)
- Current again flows in the same direction through the load
- Output remains positive

Thus, both halves of AC waveform contribute to DC output, increasing efficiency.

Advantages of Bridge Rectifier

- Full-wave rectification
- No center-tap needed
- Higher output voltage
- Lower ripple
- Better transformer utilization

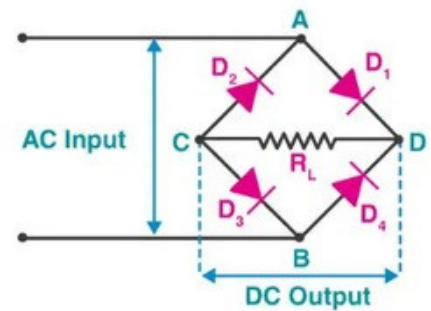


fig 1.3.1

1.4 Voltage Regulator ICs (7805, 7809, 7812)

Voltage regulation is one of the most important parts of any DC power supply. Even after AC is rectified and filtered, the resulting DC voltage is not completely stable. It usually varies between 13V to 17V, depending on load and input fluctuations. Such changes are harmful to sensitive electronic devices like microcontrollers, sensors, digital ICs, audio circuits, communication circuits, etc. This is where voltage regulator ICs come into use.

The 78XX series of regulators are among the most widely used, reliable, and cost-effective linear voltage regulators.

The "XX" in "78XX" represents the output voltage.

For example:

- 7805 → 5V regulator
- 7809 → 9V regulator
- 7812 → 12V regulator

These are three-terminal regulators with the pins:

1. Input (V_{in})
2. Ground (GND)
3. Output (V_{out})

They are extremely easy to use and provide high-quality regulated DC output.

1.4.1 Why Voltage Regulation is Needed?

After rectification and filtering, the output DC still contains:

- Ripple voltage
- High amplitude peaks
- Variation due to line voltage changes
- Load-dependent fluctuations
- Noise and transients

For example, when the mains AC supply fluctuates between 190V to 250V, the secondary AC voltage of the transformer also changes. As a result, the rectifier DC output also fluctuates. These fluctuations can damage ICs, reduce the life of components, and create malfunction in circuits.

A voltage regulator ensures:

- Constant voltage, even if input changes
- Protection from voltage spikes
- Stable operation of sensitive components
- Noise-free power supply
- Elimination of ripple and distortion

Thus, regulators are the backbone of any reliable DC supply.

1.4.2 Internal Working of Voltage Regulator IC

The 78XX series are linear series regulators.

They consist of:

- Voltage reference circuit
- Error amplifier
- Pass transistor (Series transistor)
- Short circuit protection
- Thermal shutdown system

Working Process:

- The internal reference generates a fixed reference voltage.
- This reference is compared with the output using an error amplifier.
- If the output goes higher than expected, the IC decreases the conduction of its pass transistor, reducing output.
- If the output dips, the pass transistor increases conduction, raising output.

This ensures constant output under varying conditions.

1.4.3 Types of 78XX Voltage Regulators Used

1.4.3.1 7805 – 5 Volt Regulator

- Output: 5V DC
- Used for microcontrollers, logic circuits, sensors
- Ideal for digital electronics such as Arduino, Raspberry Pi modules

1.4.3.2 7809 – 9 Volt Regulator

- Output: 9V DC
- Used for RF circuits, analog sensors, small motors, audio preamps
- Stable enough for low-noise circuits

In your project, 7809 regulates 9V output from ~15–17V input.

1.4.3.3 7812 – 12 Volt Regulator

- Output: 12V DC
- Used for relays, solenoids, LED strips, fans, amplifiers
- Highly useful for medium-power loads
- 7812 provides constant 12V for higher power components.

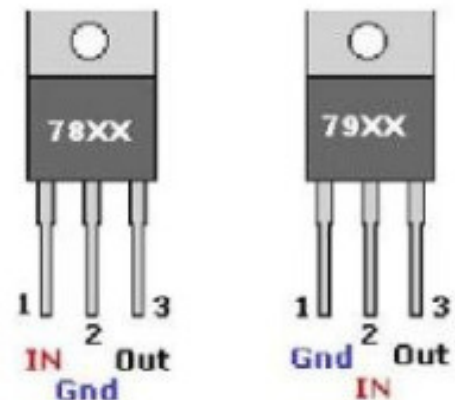


Fig 1.4.1

1.5 Filter Capacitor (1000 μF / 25V)

After the AC voltage is rectified using a bridge rectifier, the output obtained is pulsating DC. This pulsating DC is not suitable for electronic circuits because it contains a large amount of ripple—small unwanted AC component superimposed on the DC waveform. Ripple reduces efficiency, creates noise in audio circuits, causes malfunction in digital circuits, and can even damage sensitive ICs.

To remove this ripple and smooth the DC voltage, a filter capacitor is added immediately after the rectifier.

In this project, a 1000 μF , 25V electrolytic capacitor is used as the main filter capacitor.

1.5.1 Need of a Filter Capacitor (Why filtering is necessary?)

When AC is converted to DC, the output voltage goes up and down according to the AC waveform.

During every cycle:

- At the peak of the AC waveform $\rightarrow$ Output becomes high
- At the zero crossing $\rightarrow$ Output drops to zero

Without a filter capacitor, the output DC looks like a series of humps, not a straight line.

This is extremely harmful because:

- ICs require steady DC
- Motors run unevenly
- LEDs flicker
- Microcontrollers reset
- Amplifiers produce humming noise
- Sensors give incorrect readings

So the filter capacitor is the heart of smoothing and stabilizing DC supply.

1.5.2 How a Filter Capacitor Works (Charging & Discharging Process)

A large electrolytic capacitor is placed across the DC output.

It performs two main actions:

1. Capacitor Charging (at voltage peaks)

During the peak of the rectified DC waveform, the capacitor charges up to the maximum voltage level.

2. Capacitor Discharging (when voltage drops)

When the input voltage falls (during the valley of the waveform), the capacitor discharges and supplies current to the load.

This discharge keeps the voltage from dropping too low.

This continuous cycle results in:

- Reduced ripple
- Smoother output
- More stable DC supply

The capacitor essentially acts like a temporary rechargeable battery that fills the gaps between the ups and downs of the rectified waveform.

1.5.3 Why 1000 μF Value Is Used? (Deep Explanation)

The capacitor value directly affects the amount of filtering.

Higher capacitance $\rightarrow$ better smoothing.

The commonly used thumb-rule:

Where,

$$C = \frac{I}{2fV_{\text{ripple}}}$$

- I = Load current
- f = Frequency of AC $\times 2$ (full-wave rectifier) $\rightarrow 100 \text{ Hz}$
- V_{ripple} = Allowable ripple voltage



Fig 1.5.1

1.6 Resistors

A resistor is a fundamental passive electronic component used to oppose or limit the flow of electric current in a circuit. It works on the principle of electrical resistance, which is measured in ohms (Ω). Resistors help control voltage and current levels, protect sensitive components, divide voltages, set biasing conditions, and shape signal waveforms.

They are available in different types such as carbon film, metal film, wire-wound, and variable resistors, each designed for specific applications based on precision, power rating, and stability. In power supply circuits, resistors are commonly used in LED indicators, voltage dividers, and to limit current to ensure safe operation of components.

1 k Ω Resistor (Used in the Project) – Point-wise Description

Current Limiting:

The 1 k Ω resistor is connected in series with each LED to limit the amount of current flowing through it and prevent the LED from burning out.

Protects the LED:

LEDs typically require only 10–20 mA of current. Without a resistor, the regulated 9V or 12V would damage the LED. The 1 k Ω resistor ensures safe operating current.

Ensures Long Life of Components:

By restricting excessive current, the resistor increases the lifespan of both the LED and the connected regulator IC.

Stable Indication:

Using 1 k Ω provides a balanced brightness—neither too dim nor overly bright—making the power indicator visually stable and clear.

Voltage Drop Management:

The resistor drops the excess voltage that the LED cannot handle, allowing it to operate at its typical forward voltage (about 2V for red LED).

Simple and Reliable:

A 1 k Ω value is commonly used in power supply circuits because it suits most LEDs and ensures consistent performance without the need for calculations each time.

Helps Identify Voltage Output:

In your project, one LED is placed at the output of LM7809 and another at LM7812. The 1 k Ω resistors help differentiate and monitor both regulated outputs safely.

Color	Color	1st Band	2nd Band	3rd Band Multiplier	4th Band Tolerance
Black		0	0	x1 Ω	
Brown		1	1	x10 Ω	$\pm 1\%$
Red		2	2	x100 Ω	$\pm 2\%$
Orange		3	3	x1k Ω	
Yellow		4	4	x10k Ω	
Green		5	5	x100k Ω	$\pm 0.5\%$
Blue		6	6	x1M Ω	$\pm 0.25\%$
Violet		7	7	x10M Ω	$\pm 0.10\%$
Grey		8	8	x100M Ω	$\pm 0.05\%$
White		9	9	x1G Ω	
Gold				x0.1 Ω	$\pm 5\%$
Silver				x0.01 Ω	$\pm 10\%$

Fig 1.6.1

1.6 LED (Light Emitting Diode)

In this regulated power supply project, a 1.5V LED is used as a visual indicator to show the proper operation of each voltage regulator output. An LED (Light Emitting Diode) is a semiconductor device that emits visible light when an electric current flows through it. It operates on the principle of electroluminescence, where energy is released in the form of light as electrons recombine with holes inside the diode material.

The LED used in this project has a forward voltage of approximately 1.5 volts, which means it requires 1.5V across it to begin glowing. LEDs are known for their low power consumption, long life, and reliability, making them ideal for use as status indicators in power supply circuits. They typically operate at a very small forward current, usually between 10 mA to 20 mA, which ensures bright illumination without generating excess heat.

To ensure safe operation, the LED is connected in series with a 1 k Ω resistor. This resistor performs the critical role of limiting the current flowing through the LED, preventing it from burning out. Since the output voltages of the regulators (9V and 12V) are much higher than the LED's required 1.5V, the resistor drops the extra voltage and maintains a safe current level. This simple arrangement ensures that the LED glows steadily whenever the regulator outputs are active, providing a clear visual indication that the power supply is functioning correctly.

The use of the LED in this power supply enhances usability by immediately showing whether the 9V and 12V outputs are available. This is especially important during testing and troubleshooting of electronic circuits. In summary, the 1.5V LED is a small but essential component that contributes to the overall reliability, safety, and user-friendliness of the regulated power supply system.

CHAPTER 2

THEORY

A regulated power supply is one of the most essential building blocks in electronics. Every electronic

device, sensor module, microcontroller, amplifier, or communication system requires a stable DC voltage to function properly. The purpose of this project is to design a transformer-based regulated dual-voltage power supply that provides 9V and 12V regulated DC outputs. To understand the working of this system, it is important to study the theory behind each stage and each component used in the circuit.

This chapter explains the complete theoretical background of the project, including the roles of the transformer, rectifier, filter capacitor, voltage regulators (LM7809 & LM7812), resistors, and LED indicators. Each component is described in detail, along with how it contributes to the overall working of the power supply.

2.1 Working Principle of a Regulated Power Supply

- A regulated power supply converts high-voltage AC from the mains into low-voltage, stable DC. The conversion takes place through four fundamental stages:
- Step-down of AC voltage using a transformer
- Rectification of AC to DC using a bridge rectifier
- Filtering of the rectified DC using a capacitor
- Voltage Regulation using regulator ICs

Each of these stages plays a critical role in ensuring a stable output voltage. Without any one of these stages, the supply would be unable to provide clean, constant, dependable DC power.

2.2 Step-Down Transformer (12-0-12V, 1A)

The first stage of the power supply is the 12-0-12V, 1A center-tapped step-down transformer. It performs multiple functions:

2.2.1 Voltage Reduction

- The mains supply (230V AC) is too high for electronic circuits. The transformer reduces this high voltage to a safe and usable level.
- The secondary output of the transformer provides 12V AC, which is suitable for rectification.
- The center tap (0V) allows the option of generating dual supplies if required.

2.2.2 Isolation

The transformer provides galvanic isolation, meaning the primary (input) and secondary (output) are electrically separated. This isolation increases safety and prevents accidental high-voltage damage.

2.2.3 How the Transformer Works

- When AC flows through the primary winding, it produces a changing magnetic field.
- This magnetic field induces an AC voltage in the secondary winding (electromagnetic induction).
- The voltage in the secondary depends on the turn ratio of primary to secondary windings.
- In this project, the transformer ensures that the circuit receives a low-voltage AC signal that is safe for the rectifier and regulator stages.

2.3 Bridge Rectifier (Using 1N4007 Diodes)

After stepping down the AC, the next stage converts AC into DC. This is done using a full-wave bridge rectifier made with four 1N4007 diodes.

2.3.1 Need for Rectification

Electronic circuits operate on DC voltage. AC fluctuates from positive to negative, which cannot be used directly in most circuits. Therefore, the AC must be converted to DC.

2.3.3 Working of Bridge Rectifier

- There are four diodes, arranged in a diamond shape:
- During the positive AC cycle, two diodes conduct and pass current.
- During the negative AC cycle, the other two diodes conduct.
- However, in both cases, the output polarity remains the same.
- This results in full-wave rectified DC, which is stronger and more continuous.

2.3.4 Why 1N4007?

- Rated for 1000V reverse voltage
- Rated for 1A continuous current
- Suitable for transformer-based power supplies
- This makes 1N4007 the ideal choice for safely handling rectification in this project.

2.4 Filter Capacitor (Electrolytic Capacitor)

- The output of the rectifier is DC, but it is pulsating (full of ripples). Electronic circuits require smooth, steady DC.
- To smooth this pulsating DC, an electrolytic filter capacitor is used.

2.4.1 Principle of Filtering

- When rectifier pulses reach the capacitor, it charges to the peak voltage.
- When the voltage drops between pulses, the capacitor discharges slowly, supplying the stored energy to the circuit.
- This reduces voltage ripple and produces a smoother DC output.

2.4.2 Importance of the Capacitor

Without a capacitor:

- The output would fluctuate heavily
- Regulators may fail to provide stable voltage
- LEDs and loads would flicker
- A large capacitor ensures minimum ripple and feeds a steady DC to the voltage regulators.

2.4.3 Value Selection

- In most supplies:
- 1000 μ F per 1A load current is a rule of thumb
- Higher value gives smoother DC
- Thus, the filter capacitor is a vital component in the power supply chain.

2.5 Voltage Regulators (LM7809 & LM7812)

After filtering, the DC is smoother but not constant. It still changes with:

- Input AC variations
- Load changes
- Temperature changes
- Ripple voltage
- Therefore, voltage regulators are used.

2.5.1 LM7809

- Provides a fixed 9V DC output
- Requires at least 11–12V input
- Used for circuits requiring 9V supply (sensors, small modules, logic circuits)

2.5.2 LM7812

- Provides a fixed 12V DC output
- Requires at least 14V input
- Used for 12V circuits (motors, relays, amplifiers)

2.5.3 Features of 78XX regulators

- Short-circuit protection
- Thermal shutdown
- Overload protection
- Low ripple output
- Very easy to use (only 2 capacitors needed)
- These regulators ensure that the output remains stable and safe, even if the input or load fluctuates.

2.6 Indicators (LED with 1kΩ Resistor)

- Each regulator output has a 1.5V LED connected through a 1 kΩ resistor.
- This LED serves as a power ON indicator.

2.6.1 Purpose of LED

- Shows that the regulator is functioning correctly
- Simplifies troubleshooting
- Adds user-friendly visual feedback

2.6.2 Why 1kΩ Resistor

- The resistor limits the current flowing through the LED.
- Without the resistor, the LED would burn out.

2.7 Complete Working of the Circuit

- Transformer reduces 230V AC to 12-0-12V AC.
- Bridge rectifier converts AC to pulsating DC.
- Filter capacitor smooths the pulsating DC to a stable DC.
- The smoothed DC is applied to both LM7809 and LM7812 regulators.
- Regulators output steady 9V and 12V respectively.
- Each regulator output passes through 1 kΩ resistor into 1.5V LED, lighting it up.
- Thus, the circuit delivers two stable DC voltages with visual indicators.

Component	Role in Circuit / Purpose
Step-down transformer (12-0-12 V, 1 A)	Converts high mains AC (230 V) to safe low-voltage AC (~12 V) suitable for electronics.
Diodes (1N4007 × 4) in Bridge Rectifier configuration	Convert AC to pulsating DC (full-wave rectification).
Filter Capacitor (Electrolytic)	Smooths pulsating DC to reduce ripple and provide stable DC.
Voltage Regulator ICs (LM7809 & LM7812)	Provide fixed, stable DC outputs (9 V and 12 V) regardless of input/load variation.

Table 2.7.1

2.8 Summary

This chapter explained the complete theoretical background of the dual regulated power supply. Each component—from the transformer to the LED—plays a critical role in ensuring stable, reliable operation. Understanding the theory helps in the design, construction, testing, and troubleshooting of power supplies used in electronics.

CHAPTER 3

CIRCUIT DIAGRAM

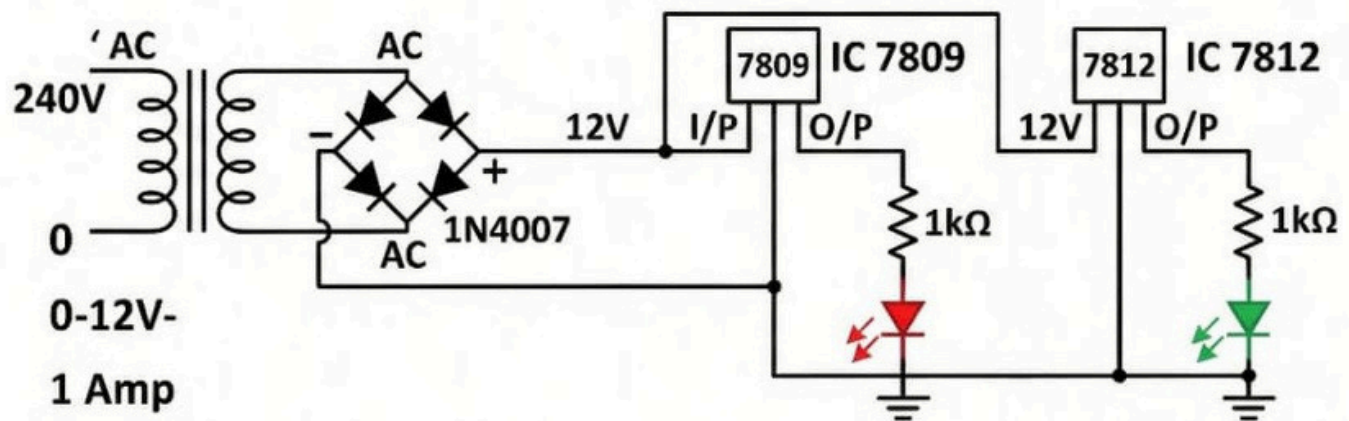
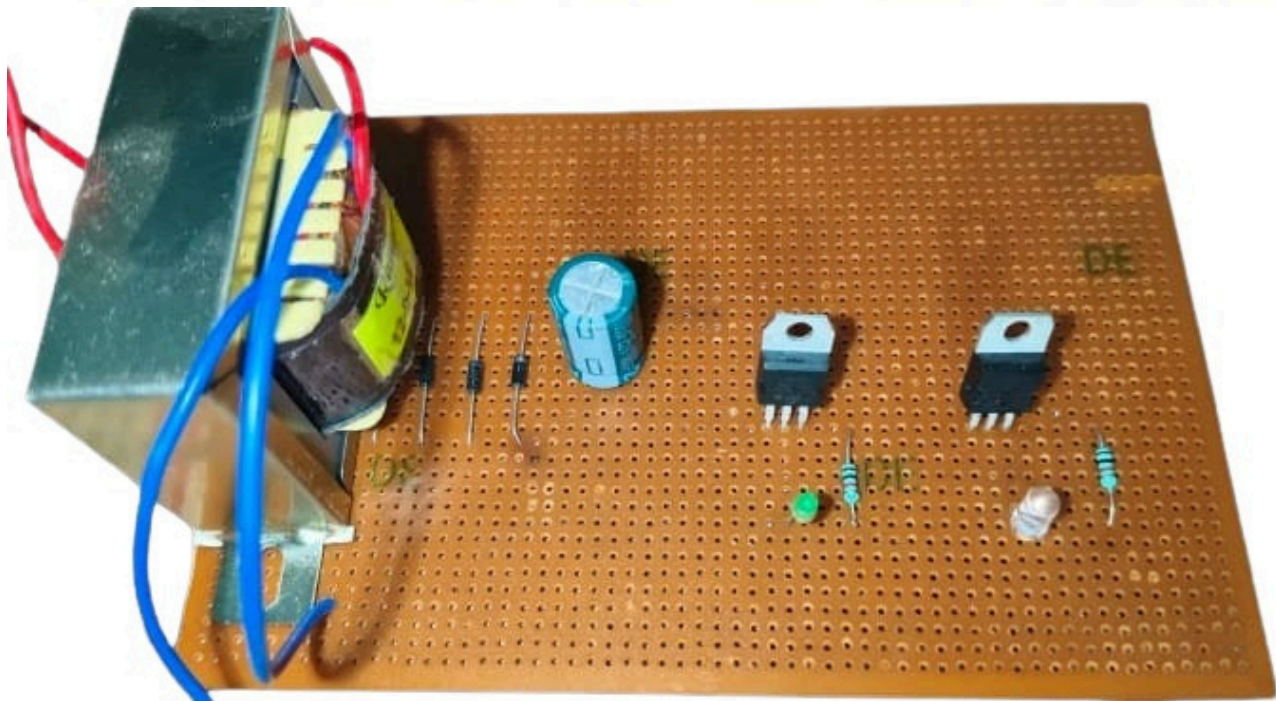


Fig 3.1



CHAPTER 4 – DERIVATION & DESIGN CALCULATIONS

4.1 Basic relationships — from AC secondary to DC

Secondary RMS (per half winding used in bridge).

$$V_{rms} = 12 \text{ V}$$

Peak of AC waveform:

$$V_{peak} = V_{rms} \times \sqrt{2} = 12 \times 1.414 \approx 16.97 \text{ V}$$

Diode drops in a bridge: during conduction current flows through two diodes (series). Assume each silicon diode forward drop $V_f \approx 0.7 \text{ V}$.

$$\text{Total diode drop} = 2 \times V_f = 1.4 \text{ V}$$

Approximate DC (no-load peak) after bridge and before filtering:

$$V_{DC, peak} \approx V_{peak} - 2V_f = 16.97 - 1.4 \approx 15.57 \text{ V}$$

Design note: choose capacitor voltage rating > this peak. Recommended: $\geq 25 \text{ V}$, safer 35 V .

4.2 Ripple frequency and ripple formula

With a full-wave bridge fed from 50 Hz mains, ripple frequency is:

$$f_{ripple} = 2 \times f_{mains} = 100 \text{ Hz}$$

For a capacitor-input filter (large electrolytic) the ripple peak-to-peak voltage (approx) is:

$$\Delta V_{pp} = I_{load} / (f_{ripple} \times C)$$

where

I_{load} = DC current drawn from the filtered bus,

C = filter capacitance (in farads),

f_{ripple} = ripple frequency.

Example 1 — with $C = 2200 \mu\text{F}$ and $I_{load} = 0.5 \text{ A}$:

$$\Delta V_{pp} = 0.5 / (100 \times 2200 \times 10^{-6}) \approx 2.27 \text{ V}$$

Minimum voltage between peaks (conservative):

$$V_{min} \approx V_{DC, peak} - \Delta V_{pp} = 15.57 - 2.27 \approx 13.30 \text{ V}$$

Example 2 — with $C = 4700 \mu\text{F}$ and $I_{load} = 0.5 \text{ A}$:

$$\Delta V_{pp} = 0.5 / (100 \times 4700 \times 10^{-6}) \approx 1.06 \text{ V}$$

$V_{min} \approx 15.57 - 1.06 \approx 14.51 \text{ V} \rightarrow$ keeps V_{min} above LM7812 dropout ($\sim 14 \text{ V}$) for 0.5 A load.

4.3 Regulator input requirements (dropout) & verification

Linear 78xx regulators require a certain headroom (dropout voltage). Typical assumptions:

LM7812 requires $\approx 2 \text{ V}$ headroom $\rightarrow V_{in(min)} \approx 12 + 2 = 14 \text{ V}$

LM7809 requires $\approx 2 \text{ V}$ headroom $\rightarrow V_{in(min)} \approx 9 + 2 = 11 \text{ V}$

Using filtered peak $\approx 15.57 \text{ V}$ and ripple, verify:

For LM7812: $V_{min} \approx 14.51 \text{ V} \geq 14 \text{ V} \rightarrow \text{OK}$

For LM7809: $V_{min} \approx 14.51 \text{ V} \geq 11 \text{ V} \rightarrow \text{OK}$

4.4 Filter capacitor value selection (practical method)

$$C \geq I_{max} / (f_{ripple} \times \Delta V_{pp, max})$$

Example: For $\Delta V_{pp, max} = 1 \text{ V}$, $I_{max} = 1 \text{ A}$, $f_{ripple} = 100 \text{ Hz}$:

$$C \geq 1 / (100 \times 1) = 0.01 \text{ F} = 10,000 \mu\text{F}$$

→ So for 1 A load and 1 V ripple target you need $\approx 10,000 \mu\text{F}$.

4.5 Regulator power dissipation & heatsinking

Linear regulators dissipate the voltage drop as heat:

$$P_{\text{diss}} = (V_{\text{in}} - V_{\text{out}}) \times I_{\text{out}}$$

Example: LM7812 at 1.0 A:

$$P = (15.57 - 12) \times 1.0 \approx 3.57 \text{ W}$$

Example: LM7809 at 1.0 A:

$$P = (15.57 - 9) \times 1.0 \approx 6.57 \text{ W}$$

Rule: dissipation $> \sim 1\text{--}2 \text{ W}$ → use heatsink.

4.6 Diode (bridge) stresses & selection

Average diode power: $P_{D,\text{avg}} \approx V_f \times I_{\text{avg}}$

Example: $I_{\text{avg}} = 1 \text{ A}$, $V_f = 0.7 \text{ V}$ → $P \approx 0.7 \text{ W}$

Peak surge currents occur during capacitor charging pulses → diode selection must handle surge.

4.7 LED resistor derivation

LED forward voltage $V_f \approx 1.5 \text{ V}$, desired current $I_{\text{LED}} \approx 10 \text{ mA}$

$$R = (V_{\text{reg}} - V_f) / I_{\text{LED}}$$

For 12 V rail: $R = (12 - 1.5) / 0.01 = 1050 \Omega \approx 1 \text{ k}\Omega$

For 9 V rail: $R = (9 - 1.5) / 0.01 = 750 \Omega \rightarrow 1 \text{ k}\Omega$ gives 7.5 mA (safe)

Power in resistor: $P = (V_{\text{reg}} - V_f) \times I_{\text{LED}}$

Example: 12 V rail → $P \approx 0.105 \text{ W} \rightarrow \frac{1}{4} \text{ W}$ resistor OK

4.8 Transformer VA check

Transformer rating: 12 V, 1 A → 12 VA

Maximum continuous DC load $\leq 1 \text{ A}$ to prevent heating and voltage drop.

CHAPTER 5 – DESCRIPTION OF COMPONENTS

The success of a regulated powersupplyliesin the careful selection and integration of its components. Each part of the circuit plays a unique role in converting AC mains voltage into a stable, low-ripple DC voltage. This project uses a dual-output regulated supply with 9V and 12V outputs, and the components have been selected to ensure efficiency, safety, and reliability. The following section provides a detailed explanation of each component, highlighting its specifications, working principle, and importance in the overall design.

5.1 Step-Down Transformer (12-0-12V, 1A)

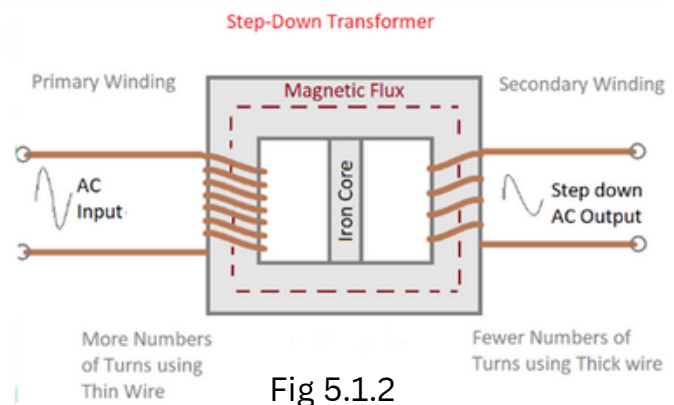
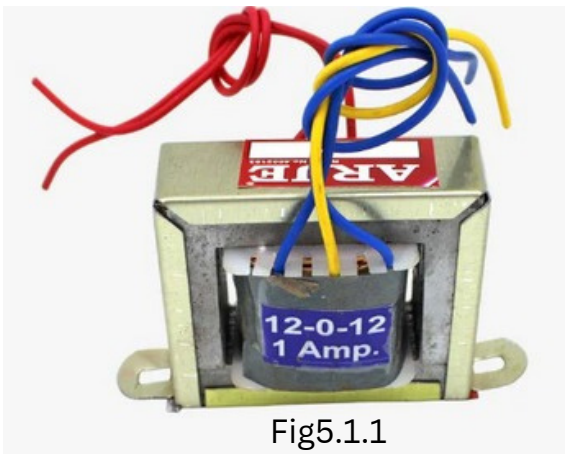
The transformer is the first and most critical component in the power supply circuit. Its primary function is to step down the high-voltage AC mains (typically 230V AC) to a lower voltage suitable for electronic circuits. In this project, a 12-0-12V, 1A center-tapped transformer is used. The center tap allows the creation of dual voltage outputs and simplifies the rectification process.

The transformer not only reduces voltage but also provides electrical isolation between the mains and the low-voltage section, which is crucial for safety. It operates on the principle of electromagnetic induction, where alternating current in the primary coil induces a voltage in the secondary coil. The transformer is designed to handle a maximum current of 1A, ensuring that the circuit can supply sufficient current to the load without overheating.

Key Features:

- Provides dual 12V AC output for versatile voltage regulation.
- Offers electrical isolation to protect the user and sensitive circuits.
- Rated for 1A load, suitable for low-to-medium power electronic devices.
- Ensures efficient energy transfer with minimal losses.

By using a transformer of appropriate rating, the circuit achieves stable input voltage for the rectifier stage, which is critical for accurate regulation.



5.2 Bridge Rectifier (1N4007 Diodes)

The AC voltage from the transformer cannot be directly used by most electronic circuits. It must be converted to DC. This is accomplished using a bridge rectifier, built from four 1N4007 diodes. These diodes are chosen for their high voltage rating of 1000V and forward current capability of 1A, which ensures reliable operation under expected load conditions.

The bridge rectifier allows full-wave rectification, meaning it uses both the positive and negative halves of the AC waveform to produce a unidirectional current. This increases efficiency compared to half-wave rectification and provides a higher average DC voltage. While the rectified output is unidirectional, it still contains ripples — voltage fluctuations that must be minimized before the voltage can be considered regulated.

Key Points:

- Converts AC to pulsating DC efficiently.
- Uses four diodes in a bridge configuration for full-wave rectification.
- Protects the circuit against reverse current.
- Forms the backbone of the DC supply before smoothing and regulation.

By carefully arranging the diodes, the bridge ensures continuous and reliable current flow to the filtering stage.

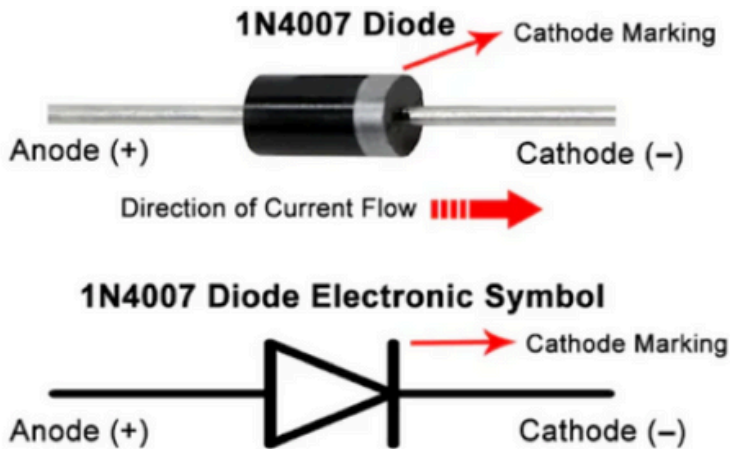


Fig 5.2.1

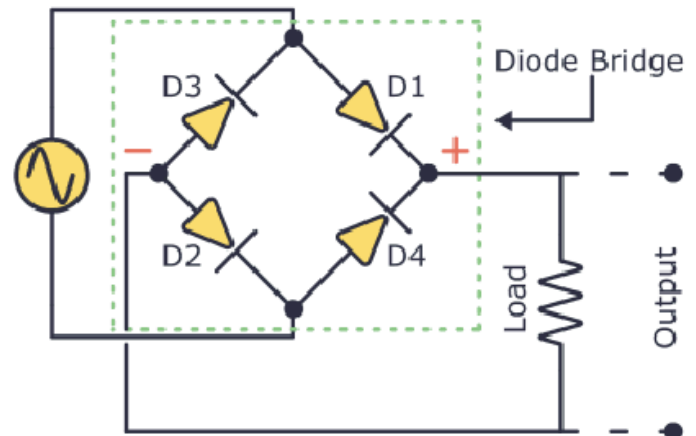


Fig 5.2.2

5.3 Filter Capacitor (1000 μ F / 25V Electrolytic)

After rectification, the DC output still contains significant ripples. To smooth this, a 1000 μ F, 25V electrolytic capacitor is used. This capacitor plays an essential role in reducing the ripple voltage and producing a steady DC level. When the rectified output is high, the capacitor charges to maximum voltage; when the voltage drops between cycles, it discharges to maintain output continuity.

The 1000 μ F value provides large charge storage ability, making it ideal for smoothing applications. The 25V rating ensures that the capacitor can withstand the peak rectified voltage (which can reach 17–18V after rectification from 12V AC). The electrolytic design gives high capacitance in a compact size.

Detailed Points:

- Capacitance: 1000 μ F, ideal for ripple smoothing.
- Voltage rating: 25V, safely above working voltage.
- Reduces ripple and stabilizes DC output.
- Works by storing and releasing charge in each AC cycle.
- Essential for proper performance of voltage regulator ICs.
- Improves voltage quality and reduces noise.



Fig 5.3.1

5.4 Voltage Regulators (LM7809 and LM7812)

Precise voltage regulation is achieved using LM7809 (9V) and LM7812 (12V) fixed linear regulators. These ICs ensure that the output voltage remains constant despite fluctuations in input voltage or load current. They are widely used due to their stability, built-in protection features, and ease of use.

Each regulator includes short-circuit protection, thermal overload protection, and safe-area compensation, making them reliable even under stressful conditions. When combined with the filter capacitor, these regulators provide near-perfect DC with very low ripple.

Detailed Points:

- LM7809 provides 9V regulated output.
- LM7812 provides 12V regulated output.
- Built-in thermal shutdown prevents overheating.
- Built-in short-circuit protection ensures safety.
- Requires minimal external components for operation.
- Provides clean, low-ripple, stable DC for the load.



Fig 5.4.1

5.5 1kΩ Resistor

The 1kΩ resistor is used primarily for limiting current to the LED indicators. LEDs typically require a very small current (10–20 mA), and without a resistor, they would burn out immediately when connected to a 9V or 12V supply. The resistor drops excess voltage and maintains safe current flow based on Ohm's Law.

The 1kΩ value is selected to ensure that the LED receives the correct current for optimal brightness without overheating or damage.

Detailed Points:

- Limits current to safe LED operating levels.
- Prevents LED burning due to excess voltage.
- Improves LED lifespan and reliability.
- Value: 1kΩ, suitable for 9V and 12V outputs.
- Ensures stable and safe LED indication.



Fig 5.5.1

5.6 1.5V Indicator LED

The LED acts as a power-on indicator for each regulator output. It lights up when the respective regulator (9V or 12V) is functioning correctly. LEDs operate at a forward voltage of around 1.5V to 2V and require a resistor in series to prevent excessive current flow. They provide immediate visual confirmation of circuit status and help in easy troubleshooting.

Detailed Points:

- Forward voltage: approximately 1.5V.
- Indicates presence of regulated output.
- Operates at low current (10–20 mA).
- Used with a 1k Ω resistor for protection.
- Helps in testing and status identification.



Fig 5.6.1

5.7 CONNECTING WIRES AND PCB

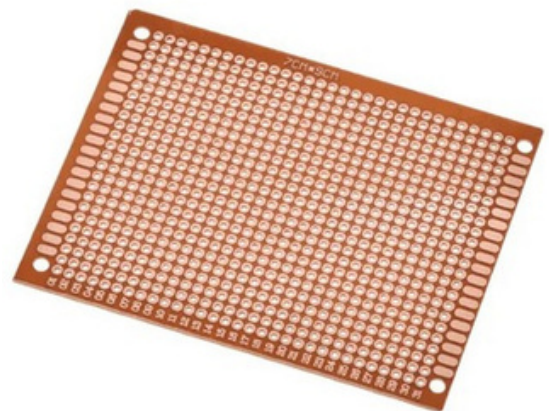
Standard insulated wires connect all components. A PCB (or breadboard) is used to mount the components neatly and ensure reliable connections.

- **Connecting Wires:** Provide connections between transformer, rectifier, capacitor, ICs, and output terminals. Proper wire thickness ensures minimal voltage drop and safety.
- **Printed Circuit Board (PCB) :** Supports and organizes components neatly. Ensures solid electrical connections and reduces circuit errors.



Connecting Wires

Fig 5.7.1



Printed Circuit Board (PCB)

Fig 5.7.2

CHAPTER 7 ADVANTAGES

ADVANTAGES OF THE REGULATED POWER SUPPLY

- Provides stable and regulated voltage outputs, ensuring that the 9V and 12V outputs remain constant even when the input voltage or load changes.
- Produces low ripple and noise because the filter capacitor smooths the DC output, making it suitable for sensitive electronic devices and microcontroller circuits.
- Offers dual voltage outputs (9V and 12V) from a single circuit, which increases flexibility and allows multiple devices to be powered simultaneously.
- Ensures electrical safety and isolation through the step-down transformer, protecting the user and connected devices from the high-voltage AC mains.
- Includes built-in protection features such as thermal shutdown and short-circuit protection provided by LM7809 and LM7812 regulators, increasing reliability.
- Uses simple and easily available components, making the entire power supply easy to build, repair, and maintain without requiring advanced tools.
- Provides long-term durability because the components used (diodes, capacitors, regulators) are robust and designed to work efficiently over extended periods.
- Reduces the chances of circuit damage by converting high AC voltage to safe DC levels suitable for common electronics.
- Gives clear visual indication through LEDs that show whether the 9V and 12V outputs are active, helping in quick verification and troubleshooting.
- Suitable for a wide range of applications, including student experiments, lab testing, hobby projects, communication modules, audio circuits, and microcontroller boards.
- Highly cost-effective since the components are inexpensive and the design is simple, making it ideal for educational and small-scale professional use.
- Offers high reliability in performance, ensuring steady output voltage, minimal heat generation, and consistent operation even under varying load conditions.

CHAPTER 8 DISADVANTAGES

DISADVANTAGES OF THE REGULATED POWER SUPPLY

Low Efficiency:

- Linear voltage regulators like LM7809 and LM7812 dissipate excess voltage as heat, making the power supply less efficient compared to modern SMPS designs.

Heat Generation:

- When supplying higher loads, the regulators may heat up significantly and sometimes require heatsinks to maintain safe operating temperatures.

Bulky and Heavy:

- The use of a 120-12V transformer increases the overall size and weight, reducing portability and making the design less compact.

Limited Current Capacity:

- The output current is restricted by the transformer and regulator ratings (typically 1A), making it unsuitable for high-power applications.

Fixed Output Voltages:

- The power supply provides only fixed 9V and 12V outputs, offering no flexibility for adjustable

Higher Cost Compared to SMPS.

- Transformer-based supplies are generally more expensive and heavier than switch-mode power supplies with similar output ratings.

CHAPTER 9 APPLICATION

APPLICATIONS OF THE REGULATED POWER SUPPLY

Electronics Laboratory Experiments:

- The power supply is widely used in schools, colleges, and training labs for powering different electronic circuits during practical experiments and project development.

Microcontroller and Embedded Systems:

- It provides stable 9V and 12V outputs needed for microcontrollers like Arduino, PIC, 8051, and

sensors, ensuring accurate performance without voltage fluctuations.

Communication and RF Modules:

- Many communication circuits such as RF transmitters, receivers, and modems require a clean and

noise-free DC supply, which this regulated power source provides.

Audio Amplifiers and Sound Circuits:

- Audio devices require low-noise DC to avoid hum in speakers. The regulated power supply offers

ripple-free voltage, ideal for audio amplifier testing.

LED and Lighting Circuits:

- LED indicators, displays, and lighting modules can be safely powered due to the stable and low-

ripple DC output from the supply.

Repair and Troubleshooting of Electronics:

- Technicians use such power supplies to test and diagnose faulty circuits by providing a safe and

controlled power source during repair work.

Battery Charging Applications:

- The 12V regulated output can be used for charging small batteries (with proper charging circuits),

offering safe and steady voltage.

Motor and Relay Control Circuits:

- Small DC motors, relays, solenoids, and control circuits often operate at 9V or 12V, making this power supply suitable for automation and hobby applications.

CHAPTER 9

CONCLUSION

The design and implementation of a transformer-based dual voltage regulated power supply (9V and 12V) successfully demonstrate the fundamental principles of AC to DC conversion, rectification, filtering, and voltage regulation. This project provides a reliable, stable, and safe DC power source that can be used for a wide range of electronic applications, making it highly valuable for both academic learning and practical use.

The use of a 12-0-12V, 1A transformer ensures proper isolation from the AC mains while stepping down the voltage to a suitable level for rectification. The 1N4007 diode bridge rectifier efficiently converts AC to pulsating DC, which is then effectively smoothed by a filter capacitor to reduce ripples. The incorporation of LM7809 and LM7812 voltage regulator ICs guarantees constant, regulated 9V and 12V outputs respectively, regardless of load variations or input fluctuations. Additionally, the inclusion of LEDs through 1 k Ω resistors provides clear visual indication for output status, making the system user-friendly and easy to inspect.

Throughout this project, it becomes clear that regulated power supplies play a crucial role in electronic systems, ensuring that sensitive components operate under safe and stable voltage conditions. Although the design has certain limitations—such as heat generation in linear regulators and limited current output it remains highly effective for low-power and medium-power applications.

Overall, the project demonstrates that a simple, cost-effective, and robust power supply can be built using easily available components, making it an excellent practical learning tool for students and hobbyists. The successful construction and testing of the dual output regulated power supply validate the theoretical concepts studied and highlight the importance of proper power management in electronic circuit design.

References

- Boylestad, R. L., & Nashelsky, L. Electronic Devices and Circuit Theory. Pearson Education, 2015.
- Datasheet – LM7809 & LM7812 Voltage Regulator ICs, Texas Instruments.
- Datasheet – 1N4007 Diode, Vishay Semiconductors.
- Datasheet – Step-Down Transformer (12-0-12V, 1A), Universal Technical Specifications.
- Online Electronics Tutorials – Power Supply Basics, Rectifiers, and Voltage Regulators.